

Abstract No.

Effects of Polystyrene Nanoplastics on Neural and Glioblastoma Stem Cells

*Marwen Ben Fdilen^{1, 2, *}, Pia Gallego¹, Elvira María Martínez¹, María de los Ángeles Marqués Torrejón¹, Clément Guillou³, Pascal Cosette³, and Olfa Masmoudi²*

¹University Jaume I, Facultat de Ciències de la Salut

²University of Tunis El Manar, Faculty of Sciences of Tunis

³University Rouen Normandy

*Presenting author

Introduction: Micro- and nanoplastics (MNPs) are emerging environmental pollutants with potential health risks, yet their impact on brain tumor progression remains unclear. Given the high proliferative capacity of glioblastoma (GBM), understanding the effect of Polystyrene nanoplastic particles (PS-NPs) on tumor interactions is critical for assessing environmental contributions to cancer biology. Cellular models such as Neural stem cells (NSCs) and glioblastoma stem cells (NPEs), which are key drivers of tumor initiation, progression, and therapeutic resistance, provide relevant cellular models for investigating these effects. **Methods:** Murine adult neural stem cells (NSCs) and glioblastoma stem cells (NPEs) were exposed to varying concentrations of polystyrene nanoplastics (100nm) for 24–72 h. Cell viability was assessed by MTT assay, and fluorescent microscopy was used to examine morphological changes. Ki67, PH3, H2AX, and active Casp3 localization were visualized by immunofluorescence with confocal microscopy. Global proteomic profiling was performed by protein extraction, SDS-PAGE separation, tryptic digestion, and HPLC–MS/MS analysis, followed by protein identification and quantitative analysis. **Results:** Exposure to PS-NPs induced dose- and time-dependent changes in cell viability. Immunofluorescence analyses revealed differential expression of Ki67, PH3, γH2AX, and cleaved caspase-3 between NPEs and NSCs, indicating that PS-NPs elicit distinct cellular responses in glioblastoma stem cells compared with healthy neural stem cells. Proteomic analysis further identified alterations in proteins associated with neurodegeneration, DNA damage, cell proliferation, and survival pathways. **Conclusions:** PS-NPs promote pro-proliferative responses in glioblastoma stem cells and may activate molecular pathways that support tumor growth and survival, while promoting neurodegeneration and death in healthy neural stem cells.

Keywords: Polystyrene nanoplastics; glioblastoma stem cells; neural stem cells; tumor growth; neurodegeneration; DNA damage; apoptosis.



Supported by SciSerTec

Web: <http://www.scisertec.de>